

Assignment 2:

USE GCP CLOUD TO CREATE A VM TO LEVERAGE AUTO
SCALING AND SECURITY

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INTRODUCTION

This report explains the step-by-step process for launching a Virtual Machine (VM) in the Google Cloud Platform (GCP), setting up auto-scaling configurations, and securing resources using firewall settings and IAM roles.

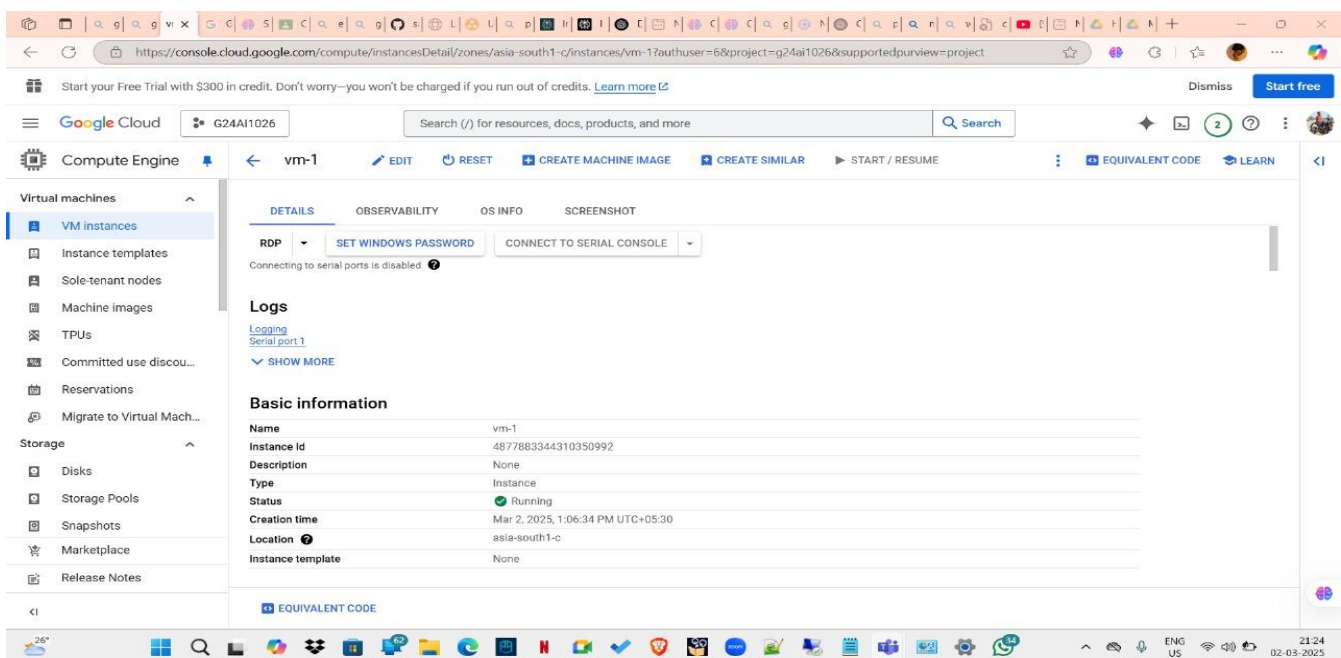
I. CREATING A VIRTUAL MACHINE IN GCP

I.1 ACCESSING THE GCP PLATFORM

- Open Google Cloud Console.
- Ensure proper permissions for VM creation.

I.2 DEPLOYING A VM INSTANCE

- Navigate to Compute Engine.
- Click **VM Instances** > **Create Instance**.
- Configure the following:
 - Machine Type: choosing E2 CPU and memory configuration.
 - Boot Disk: Select an operating system such as Windows Server
 - Firewall Rules: Enable HTTP/HTTPS traffic if required.
- Create to launch the VM instance.



2. SETTING UP AUTO-SCALING POLICIES

2.1 DEFINING AN INSTANCE TEMPLATE

- Go to Compute Engine > Instance Templates > Create Template.
- Configure the template with the same VM-01 settings.
- Click **Save**.

2.2 CONFIGURING AUTO-SCALING

- Navigate to Compute Engine > Instance Groups > Create Group.
- Select Managed Instance Group (MIG).
- Choose the instance template created earlier.
- Define Autoscaling policies:
 - Enable autoscaling.
 - Set up metrics such as CPU utilization (e.g., increase instances when CPU usage exceeds 60%).
 - Define minimum and maximum instances to ensure scalability limits as minimum as 2 and maximum as 5.
- Click Create.

The screenshot displays the Google Cloud Console interface for an auto-scaling group instance. The left sidebar shows the navigation menu with categories like Virtual machines, Storage, and Compute Engine. The main content area is titled 'auto-scaling-g...' and includes tabs for DETAILS, OBSERVABILITY, and OS INFO. Under the DETAILS tab, there are sections for Logs, Basic information, and EQUIVALENT CODE. The Basic information section contains a table with the following data:

Property	Value
Name	auto-scaling-group-It0v
Instance id	2337016121181975921
Description	None
Type	Instance
Status	Running
Creation time	Mar 2, 2025, 1:11:22 PM UTC+05:30
Location	asia-south1-c
Instance template	instance-auto-scale (Regional)

The bottom of the screen shows the Windows taskbar with various application icons and the system clock indicating 21:25 on 02-03-2025.

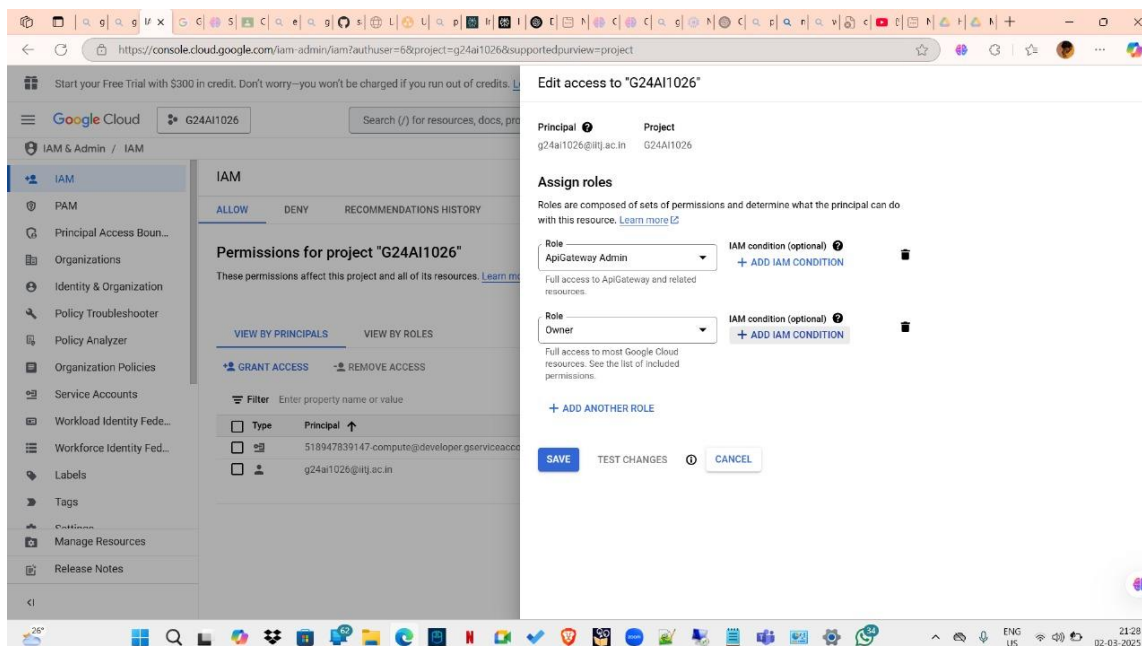
3. ENHANCING SECURITY MEASURES

3.1 MANAGING IAM PERMISSIONS

- Navigate to IAM & Admin > IAM.
- Click **Add** to grant user roles.
- Assign relevant permissions:
 - **Compute Viewer:** Read-only access.
 - **Compute Admin:** Full administrative control.
 - **Custom Roles:** Assign specific capabilities as needed.
- Click **Save**.

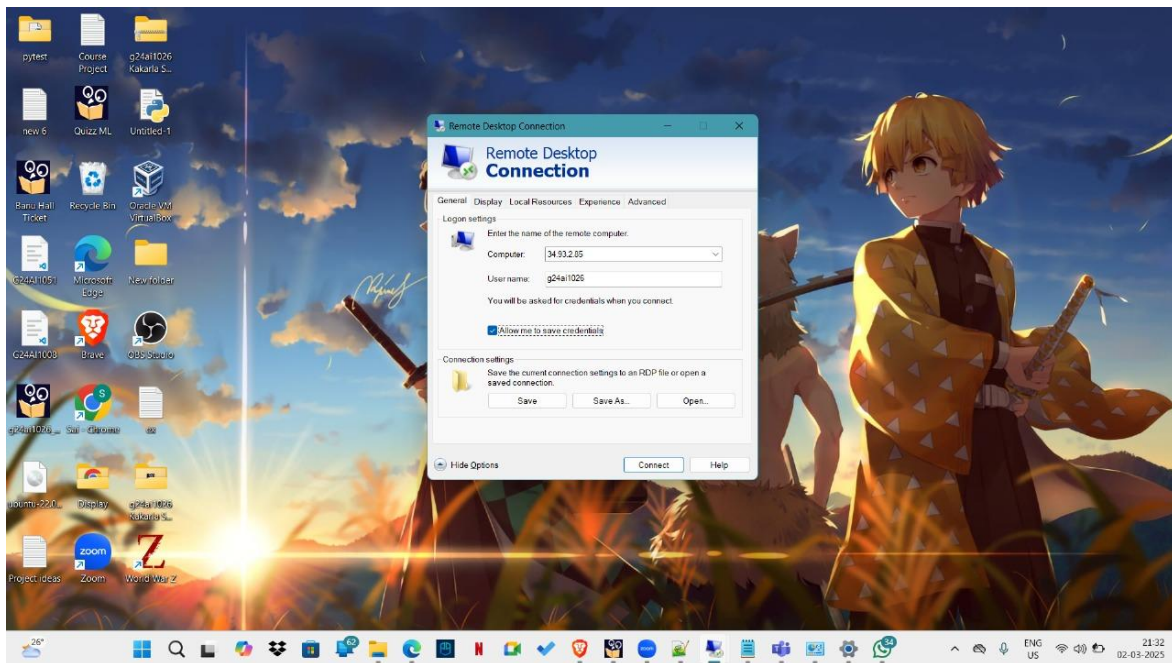
3.2 CONFIGURING FIREWALL RULES

- Navigate to VPC Network > Firewall > Create Rule.
- Configure the following:
 - **Rule Name:** Define a unique name.
 - **Traffic Type:** Choose Ingress (incoming) or Egress (outgoing).
 - **Target Instances:** Specify applicable machines.
 - **Allowed IP Ranges:** Restrict traffic accordingly.
 - **Protocols & Ports:** Allow only necessary services (e.g., TCP: 22 for SSH, TCP: 80 for HTTP).
- Click **Create** to apply security rules.



4. ACCESSING VIRTUAL MACHINE FROM PHYSICAL SYSTEM

- Setting Up remote desktop:
- Navigate to Remote desktop.
- Paste the external IP from the VM that was created early.
- Select the location pathway to share the files in the VM.
- Get the password from the GCP – VM that we created.
- And connect.



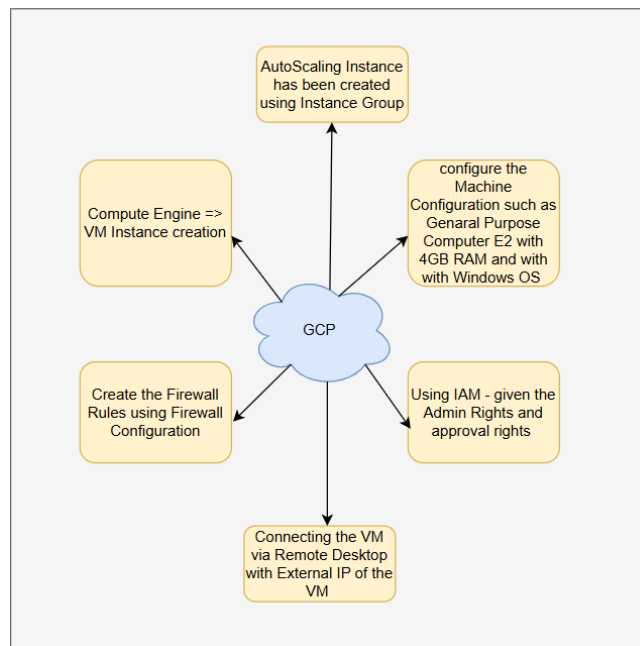
5. GITHUB LINK AND VIDEO LINK

- Git hub repo - https://github.com/saiswaroopkakarla/VCC_Assignments.git
- Video Link –
https://drive.google.com/drive/folders/1o-QWLGft9YIQGY6ka_IBGZTZqGBYHBxS?usp=drive_link

4. SYSTEM ARCHITECTURE OVERVIEW

4.1 KEY COMPONENTS

- Virtual Machine (VM): The primary compute resource.
- Managed Instance Group (MIG): Automates instance scaling.
- Load Balancer (Optional): Balances incoming traffic.
- Firewall Policies: Restricts unauthorized access.
- IAM Controls: Defines user permissions.



6. CONCLUSION

By following these steps, a virtual machine can be successfully deployed in GCP with an optimized scaling strategy and secured access control policies.

7. REFERENCES

- Compute Engine (VMs) Overview: <https://cloud.google.com/compute/docs>
- Managed Instance Groups and Auto-Scaling: <https://cloud.google.com/compute/docs/instance-groups>
- Firewall Rules in GCP: <https://cloud.google.com/vpc/docs/firewalls>
- IAM Roles and Permissions: <https://cloud.google.com/iam/docs/roles-overview>